

Normalization Assignment

Assignment: Telecom Database Normalization

Step-by-Step Questions and Answers

Scenario

A telecom company stores customer subscription data in one table.

customer_id	customer_name	phone_number	plan_name	plan_price	services	devices	tower_id	tower_location
C001	Aung Aung	09123456789	Premium	30000	Voice, SMS, Data	iPhone, Router	T01	Yangon
C002	Su Su	09987654321	Basic	10000	Voice, SMS	Samsung	T02	Mandalay
C003	Kyaw Kyaw	09777777777	Premium	30000	Voice, SMS, Data	Router	T01	Yangon

Part A: Questions and Answers

Q1. What normalization problems exist in the original telecom table?

Q2. Convert the telecom table into 1NF.

Q3. What is the main issue after converting to 1NF?

Q4. Convert the 1NF table into 2NF.

Q5. What issue still exists after 2NF?

Q6. Convert the 2NF design into 3NF.

Q7. Is the telecom database now in BCNF?

Q8. What 4NF issue can exist in telecom data?

Q9. What is a possible 5NF issue in telecom data?

Q10. What is the final normalized telecom database design?